THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 2: **Polynomials** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The degree of the polynomial $4x^4 + 0x^3 + 0x^5 + 5x + 7$ is: 1

(a) 4 **(b)** 5 **(c)** 3 **(d)** 7

2. The zeroes of the polynomial $x^2 - 3x - 10$ are: 1

(a) $5, -2$ **(b)** $-5, 2$ **(c)** $10, -1$ **(d)** $-10, 1$

3. The graph of $y = p(x)$ is a parabola opening downwards that lies completely below the x-axis. The number of zeroes of $p(x)$ is: 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3

4. If α, β are the zeroes of $x^2 + 5x + 6$, then $\alpha + \beta$ equals: 1

(a) 5 **(b)** −5 **(c)** 6 **(d)** −6

5. A quadratic polynomial whose sum and product of zeroes are -3 and 2 respectively is: 1

(a) $x^2 - 3x + 2$ **(b)** $x^2 + 3x + 2$ **(c)** $x^2 + 2x - 3$ **(d)** $x^2 - 2x + 3$

6. A student states: “A quadratic polynomial can have at most three zeroes.” This statement is: 1

(a) Correct **(b)** Incorrect — it can have at most 2 zeroes
(c) Correct for some polynomials **(d)** Cannot be determined

7. If 2 is a zero of $x^2 - kx + 6$, then the value of k is: 1

(a) 5 **(b)** −5 **(c)** 6 **(d)** 3

8. For $p(x) = x^2 - 16$, Student A writes the zeroes as $4, -4$ and Student B writes them as $4, 4$. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

9. The product of the zeroes of $3x^2 - 5x - 2$ is: 1

(a) $\frac{5}{3}$ **(b)** $-\frac{2}{3}$ **(c)** $\frac{2}{3}$ **(d)** $-\frac{5}{3}$

10. If the zeroes of $ax^2 + bx + c$ are equal in magnitude but opposite in sign, then: 1

(a) $a = 0$ **(b)** $b = 0$ **(c)** $c = 0$ **(d)** $a = c$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the zeroes of $4u^2 + 8u$ and verify the relationship between the zeroes and the coefficients. 2

12. Find a quadratic polynomial whose sum and product of zeroes are $\frac{1}{4}$ and -1 respectively. 2

13. A student says that the polynomial $x^2 + 1$ has two real zeroes. Justify or refute. 2

14. If α and β are the zeroes of $x^2 - 6x + 8$, find the value of $\alpha^2 + \beta^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the zeroes of $6x^2 - 3 - 7x$ and verify the relationship between the zeroes and the coefficients. 3

16. If α and β are the zeroes of $2x^2 - 5x + 3$, find 3
(i) $\frac{1}{\alpha} + \frac{1}{\beta}$
(ii) $\alpha^2\beta + \alpha\beta^2$
(iii) $(\alpha - \beta)^2$.

17. Find a quadratic polynomial whose zeroes are $3 + \sqrt{2}$ and $3 - \sqrt{2}$. Verify your answer. 3

OR

If one zero of $p(x) = 3x^2 - 8x + 2k + 1$ is seven times the other, find the zeroes and the value of k. 3

18. Draw the graph of $p(x) = x^2 - x - 6$ for $-3 \le x \le 4$ and find the zeroes of $p(x)$ from the graph. 3

OR

Verify that $1, -1$ and -3 are the zeroes of the cubic polynomial $x^3 + 3x^2 - x - 3$ and verify the relationship between the zeroes and the coefficients. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

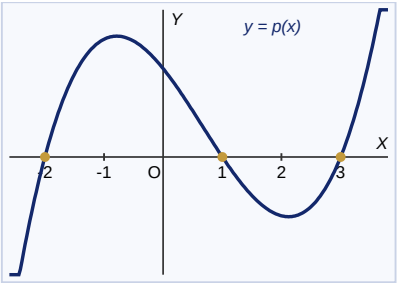
Answer in about 100–150 words each.

19. Find the zeroes of $3x^2 - x - 4$ and verify the relationship between the zeroes and coefficients. Also find a quadratic polynomial whose zeroes are the reciprocals of the zeroes of $3x^2 - x - 4$. 5

OR

Case study: The path of a ball thrown by a player is given by $h(x) = -x^2 + 4x + 5$, where h is the height (in m) and x the horizontal distance (in m). 5
(i) Find the zeroes of $h(x)$.
(ii) Which zero has a physical meaning here and what does it represent?
(iii) Verify the relationship between the zeroes and the coefficients.

20. **Case study:** The graph of a polynomial $y = p(x)$ is shown in the figure. 5



(i) How many zeroes does $p(x)$ have? Write them.
(ii) What is the least possible degree of $p(x)$?
(iii) If the leading coefficient is 1, write $p(x)$ in standard form.
(iv) Verify that the sum of the zeroes $= -\frac{b}{a}$.

OR

If α and β are the zeroes of $x^2 - 5x + k$ such that $\alpha - \beta = 1$, find k and the zeroes. Also find a quadratic polynomial whose zeroes are 2α and 2β. 5

END OF QUESTION PAPER — **Total: 40 Marks**